

Tuya: Offering comprehensive IoT solutions through an open ecosystem in cloud

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[About Tuya](#)

Tuya deploys its IoT solutions using the infrastructure of AWS Cloud to provide high stability and low latency while reducing operational costs, reaching more customers around the world.



About Tuya

Tuya Smart (NYSE: TUYA, HKEX: 2391) is a leading technology company focused on making our lives smarter. Tuya does this through offering a cloud platform that connects a range of devices via the IoT. By building interconnectivity standards, Tuya bridges the intelligent needs of brands, OEMs, developers, and retail chains across a broad range of smart devices and industries. Tuya solutions empower partners and customers by improving the value of their products while making consumers' lives more convenient through the application of

Google Cloud results

- Eliminates downtime due to operating system maintenance for higher service stability
- Helps save 10 percent of computing cost with custom VMs in Compute Engine
- Enables more efficient troubleshooting with a one-stop log query interface through Cloud Logging

As smart devices become increasingly prevalent across industries and in households, IoT service providers are exploring various ways to launch new IoT products and services easily. Depending on their IoT service providers need various

Reduction of IoT by 5

technology. Through its growing commercial SaaS business, Tuya offers intelligent business solutions for a wide range of verticals. The Company's platform is backed by industry-leading technology complete with rigorous data protection and security. Tuya partners with leading Fortune 500 companies from around the world to make things smarter, including Philips, Schneider Electric, Lenovo and many others.

Industries: Technology

Location: China

(<https://cloud.google.com/can/>), [Logging](#)



ChuanYang

About Polymericcloud

Polymericcloud is a trusted partner of Google Cloud, offering a wide range of reliable and comprehensive cloud technology solutions for companies embarking on international expansion. With a proven track record of serving over 600 million monthly active users, Polymericcloud specializes in providing expert services such as cloud technology consulting, architecture design, migration, operations, and security management. With a strong presence in key locations including China, Southeast Asia and Singapore, Polymericcloud is dedicated to

technical support to reach their development goals.

delivering the technical expertise necessary for successful overseas ventures.

With its rich experience in IoT software development, [Tuya](https://www.tuya.com) (<https://www.tuya.com>) is dedicated to offering comprehensive solutions answering to the needs of enterprise customers, ranging from consumer hardware manufacturers. Founded in 2014, the company has built an ecosystem centered around its IoT platform that supports swift software development. For organizations with limited development capability, Tuya also offers customized development solutions. Currently, it is powered by Tuya's IoT technology and has around 120,000 retail channels across the globe. According to the company's fiscal year-end report for the third quarter of 2023, as of September 30, 2023, its IoT developer platform had around 909,000 registered developers across 200+ countries and regions.

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Google Cloud (<https://cloud.google.com/>)

with a proven track record of serving over

Cloud CDN (<https://cloud.google.com/cdn/>)

Logging (<https://cloud.google.com/logging/>)

expert services such as cloud technology

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine

(<https://cloud.google.com/compute/>)

including China, Southeast Asia and

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

delivering the technical expertise

Virtual Private Cloud

(<https://cloud.google.com/vpc/>)

"Our mission is 'Build an IoT Developer Ecosystem Enable Everything to Be Smart'," says Lidong Jin, Head of Infrastructure Technology at Tuya.

"Leveraging the tools available on our developer platform, our enterprise customers can develop an IoT smart App in one minute and achieve mass production of smart devices within 15 days."

Tuya has been leveraging multiple cloud platforms to run its IoT developer platform. This way, its customers can choose the most optimal cloud platform to deploy their IoT products based on their needs. To enhance its service quality, Tuya decided

to incorporate Google Cloud (<https://cloud.google.com/>) into its multi-cloud architecture because of its high infrastructure stability. On the other hand, as Tuya has already integrated tools like Google Maps Platform (<https://mapsplatform.google.com/>) and Google Assistant (<https://assistant.google.com/>) with its IoT technology, adopting Google Cloud can pave the way for a more seamless integration with the Google ecosystem.

"We want to ensure that our enterprise customers can run their IoT products powered by our technology in the most ideal cloud environment," explains Jin. "Adding Google Cloud to our multi-cloud infrastructure means that we can provide more stable services and a wider range of cloud tools to meet various technology needs."

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*—Lidong Jin, Head of
Infrastructure Technology,
Tuya*

Enabling higher cloud infrastructure stability with lower access latency

In July 2023, Tuya completed the deployment of its IoT developer platform in Google Cloud. For customers that opted for Google Cloud as the cloud infrastructure of choice for their IoT products, the Tuya team leverages virtual machines (VMs) in Compute Engine (<https://cloud.google.com/compute>)

and containers in Google Kubernetes Engine (<https://cloud.google.com/gke>) (GKE) as servers, Virtual Private Cloud (<https://cloud.google.com/vpc>) (VPC) as network, and Cloud CDN (<https://cloud.google.com/cdn>) to deliver content to end devices.

Jin notes that since the adoption of Google Cloud, Tuya's services deployed in Google Cloud have never experienced any downtime, reaching more than 99.99 percent availability. In addition, because of the higher performance of the VMs in Compute Engine and containers in GKE, as well as the better connectivity supported by VPC, the time required for end devices to access Tuya's servers has shortened. On the other cloud platform, the average access latency was 20 to 30 milliseconds. With Google Cloud, the latency is less than 15 milliseconds.

"The higher performance and stability provided by the Google Cloud infrastructure tools like Compute Engine has helped greatly improve our service quality. Our customers can now offer more reliable and faster IoT services to their end users," says Jin.

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Reducing operations workloads while increasing troubleshooting efficiency

The more stable cloud infrastructure provided by Google Cloud also means less operations workloads to the Tuya team. There is no need for dedicated engineers to monitor services deployed in Google Cloud. Besides, Tuya employs Cloud Logging (<https://cloud.google.com/logging>) to manage the logs of the Google Cloud tools that it uses. Since Logs

Explorer

(<https://cloud.google.com/logging/docs/view/logs-explorer-interface>)

provides a one-stop interface to retrieve, view and analyze all the log data, the Tuya team can quickly query its logs when troubleshooting.

"By offering a centralized interface, Cloud Logging enables us to query our log info more efficiently. On other cloud platforms, we need to go to the interface of each cloud tool to access log info, which makes troubleshooting more complicated," notes Jin.

Saving infrastructure costs with flexible cloud resources

For the Tuya team, another advantage of using Google Cloud is that it enables the flexible use of cloud resources. As IoT services require frequent info transferring, Tuya needs VMs with high memory. On other cloud platforms that only provide VMs with fixed memory and CPU specifications, the company has to buy more CPUs than needed. On Google Cloud, the custom VMs in Compute Engine allow Tuya to purchase the exact resources needed and save 10 percent on computing cost.

Tuya normally pays additional costs on other cloud platforms, where only the standard storage class is available. By leveraging the coldline storage class of Cloud Storage (<https://cloud.google.com/storage>), Jin predicts that Tuya will be able to lower its storage cost by 20 to 30 percent in the long run.

Seamless cloud migration empowered by Google Cloud and Polymericcloud

Tuya has been collaborating with Google Cloud partner Polymericcloud to ensure the high efficiency and smoothness of its cloud migration. Throughout the process, Polymericcloud provides professional guidance and support covering cloud migration consultation, planning and implementation. Notably, the migration tool enabling real-time synchronization of object storage developed by Polymericcloud helps ensure the continuity and stability of Tuya's business.

Expanding AI applications and broadening customer reach with Google Cloud

Tuya plans to leverage the artificial intelligence (AI) technology of Google Cloud to expand the range of its IoT solutions powered by AI, such as smart cameras with more advanced recognition features and digital humans. The company will also put its IoT solutions on [Google Cloud Marketplace](https://cloud.google.com/marketplace) (<https://cloud.google.com/marketplace>) to reach more customers across regions and industries.

"We share the same value as Google Cloud with building and supporting an open, developer-friendly tech ecosystem as our core business," Jin says.

"After leveraging the cloud infrastructure of Google Cloud to realize service and cost optimization, we believe that we can empower more innovation and serve more customers with Google Cloud."

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